

Severo Ochoa Advanced School on IA for Astrophysics

Software Installation Guide

***** We strongly recommend using Linux or macOS operative systems, although we also provide instructions to install the packages on Windows.***

Complete these installations on the computer you will bring to the school and verify that they work before you arrive. Follow the instructions for your operating system. The final page includes a checklist of the expected results.

Using commands Copy each command exactly and press Enter. Use Terminal on Linux and macOS. On Windows, use PowerShell for the Docker setup and Ubuntu under [WSL 2](#) for the CTFlearn environments.

A) Google account and Google Colab

[Google Colab](#) runs in your browser and requires no local installation. **The free tier should be sufficient for the school. IMPORTANT: You should have at least 6 GB of free space in your Google Drive for the exercises' materials**

1. Sign in to [Colab](#) with your Google account, or create an account if needed.
2. Create a new notebook, click **Connect**, and enter this in a code cell:

```
print("Colab is working!")
```

3. Click the cell's Run button. Confirm that the message appears below the cell without errors.

B) Install Docker

Linux Follow the [Docker Engine installation guide](#) for your distribution. Complete its repository setup and package installation steps.

Windows Open PowerShell as Administrator and run `wsl --install`. Restart when prompted, then open Ubuntu and complete its initial setup. If WSL is already installed, run `wsl --update`. Install [Docker Desktop for Windows](#), select the WSL 2 backend when offered, and use Linux containers.

macOS Download [Docker Desktop for Mac](#) for Apple silicon or Intel. Open the .dmg file, drag Docker into Applications, and launch it to complete setup.

Check the linked guides for system requirements. On Windows and macOS, start Docker Desktop and wait for it to be ready. Test with:

```
docker run --rm hello-world
```

Check A successful test displays "Hello from Docker!" If your Linux installation requires administrator privileges, run:

```
sudo docker run --rm hello-world
```

C) Install Git

Git is used to download and work with software repositories. Install it before Git LFS.

Linux on Ubuntu or Debian

```
sudo apt update
sudo apt install git
```

On Fedora, run `sudo dnf install git`. For other distributions, follow the [Git installation guide for Linux](#).

Windows Download and run the [Git for Windows installer](#). The default options are suitable. After installation, reopen PowerShell or launch Git Bash.

macOS Install [Homebrew](#) if needed, including the “Next steps” shown by its installer. Then install [Git](#) in Terminal:

```
brew install git
```

Verify the installation on your operating system:

```
git --version
```

Check The command displays the installed Git version.

D) Install Git LFS

[Git LFS](#) extends Git to handle large files.

Linux on Ubuntu or Debian

```
sudo apt update
sudo apt install git-lfs
```

On Fedora, run `sudo dnf install git-lfs`. If the package cannot be found, follow the [official Git LFS Linux instructions](#).

Windows Git LFS is [included in Git for Windows](#). Check with `git lfs version`. If it is unavailable, install the Windows version linked from the [Git LFS website](#) and reopen your terminal.

macOS

```
brew install git-lfs
```

On every operating system, activate Git LFS for your user account and verify it:

```
git lfs install
git lfs version
```

Check Git LFS initializes and displays its version. Run these commands as your normal user, without `sudo`, before downloading the School repositories.

Windows with Ubuntu If you will use Git inside Ubuntu under WSL 2, also follow the Ubuntu instructions for Git and Git LFS in that terminal.

E) Install Miniforge

The [CTLearn tutorial](#) uses separate Python environments. [Miniforge](#) provides both **conda** and **mamba**. If both commands already work in your chosen terminal, you can use your existing installation.

Linux Download the Linux .sh installer matching your processor from the Miniforge page.

macOS Download the macOS .sh installer for Apple silicon (arm64) or Intel (x86_64).

Windows We recommend Ubuntu under [WSL 2](#) for this tutorial. Open Ubuntu and follow the Linux installation instructions inside that terminal.

1. Open a terminal in the folder containing the downloaded installer. Run **bash** followed by its filename. For example, on Linux with an Intel or AMD 64-bit processor:

```
bash Miniforge3-Linux-x86_64.sh
```

2. Follow the prompts and allow the installer to initialize Conda for your shell. Close and reopen the terminal.
3. Check that both commands are available:

```
conda --version  
mamba --version
```

For sections F and G Use Terminal on Linux or macOS, or the Ubuntu terminal on Windows.

F) Install CTFlearn

CTLearn will be used for the gamma-ray astronomy tutorial. Following the [tutorial installation instructions](#), create a dedicated environment with Python 3.12 and llvmlite, activate it, and install CTFlearn:

```
mamba create -n ctflearn -c conda-forge python==3.12 llvmlite  
conda activate ctflearn  
python -m pip install ctflearn
```

Confirm the Python version and check that CTFlearn imports:

```
python --version  
python -c "import ctflearn; print('CTLearn imports successfully')"
```

Check that the [CTLearn training command](#) starts and displays its options:

```
ctflearn-train-model --help-all
```

Check Python reports version 3.12.x, the import succeeds, and the training command displays help.

Whenever you open a new terminal to work with CTFlearn, activate the environment first:

```
conda activate ctflearn
```

G) Install ViTables

[ViTables](#) is a graphical application for inspecting HDF5 and PyTables files. It is recommended for the CTLearn tutorial and should be installed in a separate environment.

1. Create the environment:

```
mamba create -n vitables -c conda-forge python==3.12 vitables
```

2. Launch ViTables and check that its window opens:

```
conda run --no-capture-output -n vitables vitables
```

3. To open a particular data file, append its path:

```
conda run --no-capture-output -n vitables vitables "/path/to/data.h5"
```

Replace `/path/to/data.h5` with your file location. On Windows, running this graphical application inside Ubuntu requires [WSL GUI application support](#).

Before the School checklist

Complete these checks on the computer you will bring. Tick each item when its expected result appears.

Done	Software	Expected result
☐	Colab	A notebook cell prints "Colab is working!"
☐	Docker	The hello-world test displays "Hello from Docker!"
☐	Git	git --version displays the installed version.
☐	Git LFS	Initialization succeeds and git lfs version displays a version.
☐	Miniforge	Both conda --version and mamba --version work.
☐	CTLearn	Python 3.12.x is active, CTLearn imports, and training help opens.
☐	ViTables	The application window opens successfully.

If you need help

Contact the School Organizing Team before the School with your operating system and version, the terminal you used, the command you ran, and the complete error message. A screenshot can also help us identify the problem.